Finding report: How large must a program be to tell graded FairBots apart?

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1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL: p means "the opponent cooperates with me" and q "the opponent cooperates with DefectBot". Graded FairBots are $F_k = \Box^k p$ (F_1 is FairBot). They are all unexploitable and cooperate with themselves and with each other, and a LessWrong discussion [5] lists them among the agents that share these properties with FairBot without being FairBot. We take guarded formulas built from p, q, $\bot$, $\neg$, $\wedge$, $\vee$, $\Box$ and measure size as the number of nodes. An opponent *separates* two agents if it gets a different outcome against them.

Result

- **Smallest separators.** Every opponent that separates F_k from F_{k+1} has size at least $k+5$, and the only one of size $k+5$ is $S_k = \Box(\neg q \vee \Box^k \bot)$ (up to the order of the disjuncts): it cooperates with F_k and mutually defects with F_{k+1}. Without q the minimum is $k+6$, attained by $T_k = \neg\Box(\neg p \vee \Box^k \bot)$.
- **Depth.** Every opponent of modal depth at most k gets the same outcome against F_k and F_{k+1}, so a separator needs depth at least $k+1$; S_k and T_k show this is tight.
- **Ingredients.** An opponent with no negation, or with no $\wedge/\vee$, gets the same outcome against every graded FairBot; without q the same holds for every boxed opponent and every opponent with at most one negation and at most one binary connective.

Why it matters

Graded FairBots look alike: each is unexploitable, cooperates with itself and with every other graded FairBot. These results say exactly how complex a

program must be to tell two neighbouring ones apart, and that the cost grows linearly with the grade.

Proved vs checked

All statements are proved for every k in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [4], cited, not formalised (see also [3]). The uniqueness of T_k among q-free separators of size $k+6$ is proved only for $k \leq 22$, by exhaustive search; the lower bound $k+6$ holds for all k.

Novelty statement

We believe this result is new. Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire, Christiano, Fallenstein, Herreshoff, Yudkowsky, "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic" (arXiv 1401.5577): unexploitability (Section 2), rank and Theorem 4.10 (Section 4), WaitFairBot_K and the obstacle theorem (Section 5).
- LessWrong, "Is there only one FairBot?" (transhumanist_atom_understander, 29 Aug 2026) and comments: Zac Hatfield-Dodds shows the uniqueness conjecture holds up to modal depth 1 for rank-0 agents and fails from depth 2 (Box Box p; FairBot over PA+1 = Box(p or Box False); Box^k p for k >= 3); PrudentBot separates Box Box p from FairBot; the author notes Box Box p is not provably unexploitable in PA. No enumeration or counts.
- LessWrong / Alignment Forum wiki "Modal combat": DefectBot, CooperateBot, FairBot, PrudentBot, CliqueBot, TrollBot; no uniqueness result, no enumeration.
- LaVictoire et al. 2014 (MIPC); comment thread on "Robust Cooperation in the Prisoner's Dilemma" (LessWrong, on instability of FairBot equilibria); Hobson, "Yet More Modal Combat" (2021); Critch, parametric bounded Löb (arXiv 1602.04184); Critch, Dennis & Russell (arXiv 2208.07006); MIRI provability repository (github.com/machine-intelligence/provability, a modal-agent evaluator, no published census).
- Out of scope by direction: probabilistic blends (Oesterheld arXiv 2211.05057; Cooper et al. arXiv 2412.14570).
- Web searches for deterministic blends, behavioural distance and classifications of modal agents (2026-10-20/21); none means not found in this search.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result extends Blackburn, de Rijke & Venema, Modal Logic (2001); "Is there only one FairBot?" (LessWrong, 2026). Our prior-art search (2026-10-21) covered:

- Barasz, LaVictoire, Christiano, Fallenstein, Herreshoff, Yudkowsky, "Robust Cooperation in the Prisoner's Dilemma: Program Equilibrium via Provability Logic" (arXiv 1401.5577): unexploitability (Section 2), rank and Theorem 4.10 (Section 4), WaitFairBot_K and the obstacle theorem (Section 5).
- LessWrong, "Is there only one FairBot?" (transhumanist_atom_understander, 29 Aug 2026) and comments: Zac Hatfield-Dodds shows the uniqueness conjecture holds up to modal depth 1 for rank-0 agents and fails from depth 2 (Box Box p; FairBot over PA+1 = Box(p or Box False); Box^k p for k >= 3); PrudentBot separates Box Box p from FairBot; the author notes Box Box p is not provably unexploitable in PA. No enumeration or counts.
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- Out of scope by direction: probabilistic blends (Oesterheld arXiv 2211.05057; Cooper et al. arXiv 2412.14570).
- Blackburn, de Rijke & Venema, Modal Logic (2001): n-bisimulation and modal depth.
- Web searches for deterministic blends, behavioural distance and classifications of modal agents (2026-10-20/21); none means not found in this search.

If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was checked by scripts that run Lean proofs; the proof and check files are included so anyone can re-run them.

2 Claims

Claim K49

For all $k \geq 1$, every guarded separator of $F_k = \Box^k p$ and F_{k+1} has size $\geq k+5$, and every size-$(k+5)$ one is $S_k = \Box(\neg q \vee \Box^k \bot)$ up to disjunct order (CC with F_k, DD with F_{k+1}).

Status: derived ů author check only.

Verified.

- `checks/K49-c046_fk_sep.py`: Lean 4 core sizes, sk_ev, sep_coop (j<=k CC all heights), sep_defect (j>k), fk_fk, separator (all k; SHA-pinned, no sorryAx, 4 mutations rejected); exact engine over guarded classes of size <=11: k=1..6 no separator of size <=k+4, unique S_k at size k+5, k=7,8 none <=11; literal evaluator S_k vs F_j CC iff j<=k for k<=10, j<=12 — pass (Lean-proved, no sorryAx)

- `checks/K49-c046_fk_sep13.py`: guarded class counts 1..13 (36062, 100823 at 12, 13); exact engine k=1..8: no separator of F_k/F_{k+1} of size <=k+4, unique machine class S_k at size k+5; same Lean theorems as c046_fk_sep.py (sizes, sep_coop, sep_defect, fk_fk, separator; no sorryAx; 4 mutations rejected); literal evaluator S_k vs F_j CC iff j<=k — pass (Lean-proved, no sorryAx)

- `checks/K49-c046_fk_sep14.py`: guarded class counts 1..14 (100823 at 13, 279115 at 14; guarded-only products at top size); exact engine k=1..9: no F_k/F_{k+1} separator of size <=k+4, unique machine class S_k at size k+5; Lean sizes, sep_coop, sep_defect, fk_fk, separator (no sorryAx, 4 mutations rejected); literal evaluator S_k vs F_j CC iff j<=k — pass (Lean-proved, no sorryAx)

- `checks/K49-c046_fk_size40.py`: every reduced guarded formula of depth >=k+1: p,q sizes k+2..k+5 for k=1..40 (803710 solved), p-only sizes k+2..k+6 for k=1..22 (498344): no separator below k+5 / k+6, unique reduced separator S_k / T_k; full all-depth enumeration k<=2 reproduces minima without K58; bitmask games = naive literal evaluator on 1112 games — pass

- `checks/K49-c046_fk_depth.py`: K58 Lean same_outcome: no guarded opponent of modal depth <=k separates F_k/F_{k+1}, for every k — pass

- `checks/K49-c046_fk_shape.py`: Lean 4 core (SHA-pinned, includes K58 development): negfree_same (no negation: same outcome vs all F_j,F_j'), nobin_same (no &/|: same outcome vs all F_j,F_j', j,j'>=1), size_bound (k>=1: separator has ~, binary node, md>=k+1, size>=k+5), sk_size; no sorryAx; 4 mutations rejected; literal evaluator: 4333 such formulas of

size<=8 same vs F_1..F_5, all 414 separator cells (size<=9, k<=4) have ~ and a binary node — pass (Lean-proved, no sorryAx)

- `checks/K49-c046_fk_unique.py`: Lean 4 core unique_sep (all k>=1: guarded size-(k+5) separator of F_k/F_{k+1} is S_k up to disjunct order), code_same, shapeA/B/C, size_bound, sk_codes; SHA-pinned; no sorryAx/ofReduceBool; 4 mutations rejected; literal evaluator S_k k<=10 and exhaustive unreduced guarded syntax of size 6,7: only S_1,S_2 (both orders) separate — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K56

For all $k \geq 1$, every q-free guarded separator of $F_k = \Box^k p$ and F_{k+1} has size $\geq k+6$, attained by $T_k = \neg\Box(\neg p \vee \Box^k \bot)$ (DD with F_k, exploited by F_{k+1}); for $k \leq 22$ every size-$(k+6)$ one is T_k up to disjunct order.

Status: derived ů author check only.

Verified.

- `checks/K56-c046_fk_psep17.py`: p-only guarded class counts 1..17 (K51 list to 15, then 35721, 74728); exact engine k=1..11: no p-only F_k/F_{k+1} separator of size <=k+5, unique machine class T_k at size k+6 with F_k DD and F_{k+1} (D,C); k=12 none <=17; Lean tsize, tk_ev, t_low, t_high, psep for all k (no sorryAx, 4 mutations rejected); literal evaluator k<=12, j<=14 — pass (Lean-proved, no sorryAx)

- `checks/K49-c046_fk_size40.py`: every reduced guarded formula of depth >=k+1: p,q sizes k+2..k+5 for k=1..40 (803710 solved), p-only sizes k+2..k+6 for k=1..22 (498344): no separator below k+5 / k+6, unique reduced separator S_k / T_k; full all-depth enumeration k<=2 reproduces minima without K58; bitmask games = naive literal evaluator on 1112 games — pass

- `checks/K49-c046_fk_depth.py`: K58 Lean same_outcome: no guarded opponent of modal depth <=k separates F_k/F_{k+1}, for every k — pass

- `checks/K56-c046_fk_psize.py`: Lean 4 core (SHA-pinned, includes K58 and K49 shape developments): stab_box, stab_top, pone_same, one_one_same (q-free, <=1 neg and <=1 binary: same outcome vs all F_j,F_j', j,j'>=1), psize_bound (q-free separator of F_k,F_{k+1}: size >= k+6, all k>=1), tk_size; no sorryAx; 4 mutations rejected; literal evaluator 8890 small and 40758 boxed q-free formulas same vs F_1..F_6, T_k DD/CD k<=6 — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K58

For every $k \geq 1$, every guarded opponent O in p, q of modal depth $\leq k$ gets the same eventual outcome against $F_k = \Box^k p$ and F_{k+1}; so every separator has depth $\geq k+1$ (size $\geq k+2$), tight for S_k (K49) and T_k (K56).

Status: derived ů author check only.

Verified.

- `checks/K49-c046_fk_depth.py`: Lean 4 core (SHA-pinned) same_outcome for all k and guarded O with md<=k, separator_bound (md>=k+1, size>=k+2), sk_depth=k+1; no sorryAx; 5 mutations rejected; exact engine k=1..6: no depth<=k class of size<=12 separates, depth k+1 control finds S_k; literal evaluator 3200 random depth<=k opponents, exact K40 horizon — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Claim K59

For guarded O in p, q with no negation, or with no $\wedge/\vee$, and all $j, j' \geq 1$: O gets the same eventual outcome against $F_j = \Box^j p$ and $F_{j'}$ (Lean, height semantics).

Status: derived ů author check only.

Verified.

- `checks/K49-c046_fk_shape.py`: Lean 4 core (SHA-pinned, includes K58 development): negfree_same (no negation: same outcome vs all F_j,F_j'), nobin_same (no &/|: same outcome vs all F_j,F_j', j,j'>=1), size_bound (k>=1: separator has ~, binary node, md>=k+1, size>=k+5), sk_size; no sorryAx; 4 mutations rejected; literal evaluator: 4333 such formulas of size<=8 same vs F_1..F_5, all 414 separator cells (size<=9, k<=4) have ~ and a binary node — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K60

In L without q (guarded), every boxed opponent $\Box W$, and every opponent with at most one $\neg$ and at most one $\wedge/\vee$, gets the same eventual outcome against $F_j = \Box^j p$ and $F_{j'}$ for all $j, j' \geq 1$ (Lean, height semantics).

Status: derived ů author check only.

Verified.

- `checks/K56-c046_fk_psize.py`: Lean 4 core (SHA-pinned): stab_box, stab_top, pone_same (q-free O boxed or with top-level boxes negation-free or binary-free: same outcome vs all F_j,F_j', j,j'>=1), one_one_same (<=1 neg, <=1 binary), psize_bound, tk_size; no sorryAx; 4 mutations rejected; literal evaluator 8890 small and 40758 boxed q-free formulas same vs F_1..F_6 — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261020-1600`, claims K49, K56, K58, K59, K60, run-folder commit `e3d59e30bce37b58223f480d4f0d5a79bbff59ce`, published 2026-10-21.

3 Discussion

The depth bound uses the standard back-and-forth (n-bisimulation) method of modal logic [2]; the new part is its application to games between modal agents. The results concern separating graded FairBots from each other; how large a separator must be for other pairs of agents (for example the gate-family members of earlier findings) is open, as is a numerical bound on separator size at fixed modal depth.

4 How to verify

Each finding has its check scripts in `claims/graded-fairbot-separators/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[3] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[4] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[5] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.